

PRAMISH PANDEY

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EDUCATION

University of Southern Mississippi

Expected May 2027

B.S. Computer Science | GPA: 4.00/4.00

Relevant Coursework: Machine Learning, Introduction to AI, Database Management Systems, Algorithms, Linear Algebra, Probability & Statistics

PUBLICATIONS

- Pandey, P., et al. "A Centralized Data Platform Enabling Collaborative, Sample-Level Access for Polymer Morphology Research" (in preparation), School of Polymer Science & Engineering, University of Southern Mississippi.

RESEARCH EXPERIENCE

AI4ALL Ignite Fellow — AI4ALL

Jan 2026 – Present

- Engineered an end-to-end Python machine learning pipeline to process 24.9 million CDC Natality records, optimizing XGBoost and deep learning architectures (TabNet) on a 3-million-row stratified sample to predict preterm births.
- Audited and mitigated bias across 3 targeted fairness metrics using SHAP and g-computation, testing disparity hypotheses with bounded confidence intervals, and deploying a Streamlit dashboard for responsible AI auditing.

Data Engineer & Full-Stack Developer — School of Polymer Science & Engineering, USM

Sept 2025 – Present

- Designed and deployed a centralized Polymer Morphology data platform (Django, React, Google Cloud Run, Firebase Hosting) that gives collaborators shared, sample-level access to run experiments against a common database, cutting research data sorting and retrieval time by 90%.
- Built Python data pipelines and Google Cloud Storage integrations to ingest, annotate, and synthesize 50,000+ raw and unorganized research files across 20+ active laboratory studies; the resulting infrastructure is the basis of a manuscript currently in preparation.

AI Innovation Hub Fellow (Backend Developer) — MS Dept. of Information Technology Services

Feb 2026 – May 2026

- Architected a serverless AWS Step Functions pipeline orchestrating API Gateway and Lambda to automate PDF transcript processing, extracting 16 structured academic fields via Amazon Textract and Bedrock Nova Pro.
- Engineered an 18-rule deterministic Python verification engine combined with Bedrock AI analysis to evaluate synthetic records, automatically flagging academic anomalies and fraud indicators for human review.
- Developed a React single-page application with append-only audit logging for AI decisions and tailored a validated proof of concept for the Mississippi State Board of Nursing by assessing their needs in weekly standup meetings.

SELECTED PROJECTS

Igor Copilot — AI Orchestration Framework

Python, FastAPI, Gemini API, pywin32, Docker

- Built a FastAPI bridge integrating the Gemini API with Windows COM (pywin32) to automate and programmatically control WaveMetrics Igor Pro experimental software.
- Containerized the backend using Docker and enforced strict static typing and asynchronous testing (pytest-asyncio) to ensure thread-safe desktop operations.

Digital Galatea — Emotion-Aware Conversational AI

Python, Flask, Gemini API, NLTK

- Engineered a conversational AI web application integrating the Google Gemini API with Azure Text Analytics and NLTK VADER to process user sentiment and dynamically simulate 5 core emotional states.
- Constructed custom RESTful endpoints to continuously poll sentiment data from a Python (Flask) backend and synchronize visual avatar states in real time on a JavaScript frontend.

PolyVision — Polymer Image Analysis Software

C++, Qt 6, OpenCV, CMake

- Engineered a cross-platform desktop application using C++ and the Qt 6 Framework to generate, analyze, and visualize synthetic polymer images, reducing manual laboratory analysis time by 85%.
- Implemented Sobel edge detection and generative noise algorithms (Voronoi, Perlin) to batch-process 500+ high-resolution scans.

LEADERSHIP & ENTREPRENEURSHIP

Co-Founder — Clean-Slate, Guided Expungement Platform

Jan 2026 – Present

- Co-founded Clean-Slate, an automated legal-document platform that guides eligible individuals through the criminal record expungement process.
- Led technical and non-technical stakeholder presentations and raised \$1,500 in non-dilutive funding, including \$1,000 from the Hult Prize Campus Final and \$500 from Checkpoint for product R&D.

Treasurer — USM Entrepreneurship Society

Fall 2026 – Present

- Manage the organization's budget and financial planning in support of student entrepreneurship programming and events.

HONORS & AWARDS

- Academic Excellence Scholarship, University of Southern Mississippi — competitive, merit-based award covering full tuition for four years (non-resident tuition equivalent); awarded 2023 for academic performance (ACT 30+, Core GPA 3.0+).
- President's List, University of Southern Mississippi — 5 consecutive semesters.

TECHNICAL SKILLS

Languages: Python, TypeScript, JavaScript, C++, SQL, Go

AI/ML: PyTorch, scikit-learn, XGBoost, Hugging Face, LangChain, SHAP, OpenCV, NLTK, pandas, NumPy, LLM Evaluation

Web Engineering & DevOps: React, Next.js, Django, FastAPI, PostgreSQL, AWS (EC2, Lambda, Textract, Step Functions, API Gateway), GCP (Cloud Run), Firebase, Docker, Git, Agile/Scrum, CI/CD, REST APIs